

UT San Antonio

EE 4463, EE 5573, CS 6243

Homework Assignment 1

Unsupervised Machine Learning: Dimensionality Reduction with SVD

Submission and Report Requirements

- **Submission:** Submit one PDF report through Canvas by the posted deadline.
- **Teams:** Complete the assignment with a teammate of your choice; teams may consist of no more than two students. Each team member must individually submit an identical copy of the team's report. Contributions are expected to be approximately equal. Include a brief contribution statement identifying each member's substantive contributions. The instructor may require either or both team members to complete an individual oral examination and may assign different grades based on each student's demonstrated understanding and contribution.
- **Format:** Reports must be typed; handwritten work will not be graded. Number and caption every figure. Number every displayed equation that you refer to in the text. Place all requested code in an appendix entitled *Code*; code must be readable and sufficiently commented to reproduce the results.
- **Figures:** Every figure must have a descriptive caption that states its main conclusion. If information essential to interpreting a figure is not contained in its caption, provide an accompanying textual description in the report.
- **Reproducibility:** State the software environment, package versions, and all random seeds used. For Problem 6, also state the pseudorandom-number generator algorithm used.
- **References:** Cite all external sources in a reference list at the end of your report.
- **Code Appendix:** Present all code that produces your results in a dedicated appendix at the end of your report.
- **Originality:** Submitted work must reflect your team's own intellectual contribution. Do not outsource this assignment to any external human or artificial-intelligence system, including ChatGPT.
- **Grading:** Correctness, completeness, and clarity are weighted equally.

Notation and Preprocessing Conventions

All image vectors are columns. For vectors, use the Euclidean norm $\|\cdot\|_2$; for matrices, use the Frobenius norm $\|\cdot\|_F$, unless another norm is explicitly specified. For $p \geq 1$, define

$$\|v\|_p = \left(\sum_i |v_i|^p \right)^{1/p}.$$

Let $\mathbf{1}$ denote an all-ones vector of the dimension implied by context.

The required preprocessing is *mean-centering only*. Do not standardize individual pixel locations to unit variance, normalize individual images to unit norm, or rescale pixel intensities. Retain pixel intensities in the original range $[0, 255]$.

Convert retained MNIST pixel values to `float32` as specified in Problem 0. Immediately thereafter, convert X to `float64` before computing μ and $\tilde{X}$. Perform all subsequent computations, including SVDs, pseudoinverse calculations, norm calculations, numerical rank determinations, and numerical verifications, in `float64`. Use the `float64` machine epsilon $\varepsilon_{\text{mach}}$ in any numerical-rank tolerance that does not explicitly specify a different tolerance.

Problems

Problem 0 (0 points): Data Preparation

Use the *training split* of the MNIST dataset available through `tensorflow.keras.datasets.mnist`. Retain only images whose labels are 0, 1, or 9, preserving their original order. Let N be the total number of retained training images.

Each image is a 28×28 grayscale image. Convert pixel intensities to `float32`. Vectorize each image in row-major order using `image.reshape(-1)` and stack the resulting vectors as columns of

$$X = [x_1, \dots, x_N] \in \mathbb{R}^{784 \times N}.$$

Immediately convert X to `float64` before computing the quantities below. Retain the corresponding label vector so each column of X can be associated with its digit class.

Compute: Compute the global mean image

$$\mu = \frac{1}{N} X \mathbf{1} \in \mathbb{R}^{784}.$$

Compute: Form the centered data matrix

$$\tilde{X} = X - \mu \mathbf{1}^\top.$$

Verify: Numerically verify that $\tilde{X} \mathbf{1}$ is approximately zero. Report N , the dimensions of X , and the maximum absolute entry of $\tilde{X} \mathbf{1}$.

All subsequent SVD-based computations use $\tilde{X}$, except where a problem explicitly specifies otherwise.

Problem 1 (5 points): Class Centroids and Norms

For $c \in \{0, 1, 9\}$, let $\tilde{X}_c \in \mathbb{R}^{784 \times N_c}$ contain the centered samples in class c , where N_c is the number of retained images of that class.

Compute: For each class, compute its centroid in the centered coordinate system,

$$m_c = \frac{1}{N_c} \tilde{X}_c \mathbf{1} \in \mathbb{R}^{784}.$$

Compute: For each m_c , report

$$\|m_c\|_1, \quad \|m_c\|_2, \quad \|m_c\|_3.$$

Present: Report the three norms for all classes in one labeled table.

Discuss: Explain what the centered centroids represent. Discuss how the ℓ_1 , ℓ_2 , and ℓ_3 norms weight pixel deviations differently, and relate the observed values to qualitative differences among digits 0, 1, and 9.

Problem 2 (5 points): Singular Value Decomposition**Compute:** Compute an economy-size SVD of the centered data:

$$\tilde{X} = U\Sigma V^\top,$$

where $q = \min(784, N)$,

$$U \in \mathbb{R}^{784 \times q}, \quad \Sigma \in \mathbb{R}^{q \times q}, \quad V \in \mathbb{R}^{N \times q},$$

and the columns of U and V are orthonormal. The diagonal entries of Σ are the singular values.Let $r = \text{rank}(\tilde{X})$, and order the positive singular values as

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0.$$

Use and report the numerical rank tolerance

$$\tau = \max(784, N) \varepsilon_{\text{mach}} \sigma_1, \quad r = \#\{i : \sigma_i > \tau\}.$$

Plot: In Fig. 1, plot σ_i versus index i , for $i = 1, \dots, r$. A logarithmic vertical axis is permitted if it improves readability.**Compute:** Define the cumulative energy ratio

$$E(K) = \frac{\sum_{i=1}^K \sigma_i^2}{\sum_{i=1}^r \sigma_i^2}, \quad K = 1, \dots, r.$$

Plot: In Fig. 2, plot $E(K)$ versus K .**Discuss:** Interpret the singular-value decay and cumulative-energy curve in terms of data variance, redundancy, and effective linear dimensionality. Report the smallest K achieving at least 90%, 95%, and 99% cumulative energy.**Problem 3 (18 points): Low-Rank Approximation**For $1 \leq K \leq r$, define

$$Q_K = [u_1, \dots, u_K] \in \mathbb{R}^{784 \times K},$$

where u_i is the i -th column of U . Define

$$K_{\text{set}} = (\{1, 51, 101, 151, 201, 251, 301, 351, 401, 451, 501, 551, 601, 651, 701, 751\} \cap \{1, \dots, r\}) \cup \{r\}.$$

Remove duplicate values and use the resulting values in increasing order. Thus, r is always included.**Compute:** For each $K \in K_{\text{set}}$, form the rank- K approximation in centered coordinates,

$$Y_K = Q_K Q_K^\top \tilde{X},$$

and its reconstruction in original image coordinates,

$$\hat{X}_K = \mu \mathbf{1}^\top + Y_K.$$

Thus, Y_K is a rank- K linear approximation of the centered data, whereas $\hat{X}_K$ is the corresponding affine reconstruction in the original image coordinates.**Compute:** Compute the direct centered approximation error

$$e_K = \|\tilde{X} - Y_K\|_F$$

for $K = 1$, $K = 100$ if $100 \leq r$, and $K = r$. The value $K = 100$, when applicable, is an additional direct-verification rank and need not be included in K_{set} . For every $K \in K_{\text{set}}$, compute the SVD-based error

$$e_K^{\text{SVD}} = \left(\sum_{i=K+1}^r \sigma_i^2 \right)^{1/2}.$$

Use the convention that an empty sum is zero; hence $e_r^{\text{SVD}} = 0$.

Verify: Confirm numerically that the direct and SVD-based errors agree at each requested direct-computation value of K . For $K = 100$, compute e_{100}^{SVD} for this verification even if $100 \notin K_{\text{set}}$.

Plot: In Fig. 3, plot e_K^{SVD} versus K for $K \in K_{\text{set}}$. Optionally overlay the directly computed values.

Discuss: Relate the error decay to the discarded singular values. Explain why Y_K is the best rank- K approximation to $\tilde{X}$ in Frobenius norm.

Problem 4 (18 points): Storage Efficiency

The low-rank reconstruction is represented as

$$\hat{X}_K = \mu \mathbf{1}^\top + Q_K Z_K, \quad Z_K = Q_K^\top \tilde{X} \in \mathbb{R}^{K \times N}.$$

Compute: For every $K \in K_{\text{set}}$, compute the number of stored scalar values required by the full matrix,

$$S_{\text{full}} = 784N,$$

and by the low-rank representation,

$$S_{\text{LR}}(K) = 784 + 784K + KN.$$

The first term stores the mean image μ . Ignore the storage required for labels, matrix dimensions, software metadata, and numerical-format overhead; count scalar values only.

Compute: Compute the storage ratio

$$\rho(K) = \frac{S_{\text{LR}}(K)}{S_{\text{full}}}.$$

Plot: In Fig. 4, plot $S_{\text{LR}}(K)$ versus K and include a horizontal reference line at S_{full} . Clearly label the vertical axis in scalar values.

Discuss: Compare approximation quality from Problem 3 with storage efficiency. Identify the values of K yielding storage savings and explain the accuracy-storage tradeoff.

Problem 5 (18 points): Low-Dimensional Geometry and Visualization

Compute: Project the centered data onto the first two left singular vectors:

$$P_2 = Q_2^\top \tilde{X} \in \mathbb{R}^{2 \times N}.$$

For each class $c \in \{0, 1, 9\}$, compute the projected centroid

$$p_c = \frac{1}{N_c} P_{2,c} \mathbf{1} \in \mathbb{R}^2,$$

where $P_{2,c}$ contains the projected samples in class c .

Plot: In Fig. 5, create a two-dimensional scatter plot of the columns of P_2 , color-coded by digit class. Include clearly distinguishable centroid markers, a legend, and labeled axes.

Compute: Select one representative image from each class using the following reproducible rule: select the first retained training image of each class in the original MNIST training-set order. State the zero-based index of each selected image in the unfiltered `tensorflow.keras.datasets.mnist` training split; valid indices range from 0 to 59,999. Do not report the image's index after filtering. For each selected image x , reconstruct it using

$$\hat{x}_K = \mu + Q_K Q_K^\top (x - \mu)$$

with $K \in \{5, 20, 100\}$. These reconstruction ranks are independent of K_{set} . If a requested value exceeds r , use $K = r$ in its place and state this in the report.

Plot: In Fig. 6, display each original image and its three reconstructions side by side. Use a common grayscale colormap and common display limits across all images. Do not clip reconstructed values before any required numerical calculation. If clipping is used solely for visualization, disclose this explicitly.

Discuss: Describe the class geometry in the two-dimensional projection and the quality of the reconstructions as K increases.

Discuss: Propose a classification rule based on projection and reconstruction error. For example, using a training subset, estimate a mean image μ_c and rank- K subspace $Q_{K,c}$ for each class and classify a test image x by

$$\hat{c}(x) = \arg \min_{c \in \{0,1,9\}} \left\| (I - Q_{K,c} Q_{K,c}^\top)(x - \mu_c) \right\|_2.$$

You are not required to implement or evaluate this classifier. Explain its rationale and identify the data required to train it.

Problem 6 (18 points): Low-Rank Denoising

Use the original uncentered data matrix X from Problem 0. Initialize your pseudorandom-number generator with seed 4463. Construct a noisy matrix $X_n \in \mathbb{R}^{784 \times N}$ by independently corrupting each pixel. For every entry X_{ij} , independently draw a corruption indicator

$$M_{ij} \sim \text{Bernoulli}(0.05).$$

If $M_{ij} = 0$, set $(X_n)_{ij} = X_{ij}$. If $M_{ij} = 1$, independently draw

$$B_{ij} \sim \text{Bernoulli}(0.5)$$

and set

$$(X_n)_{ij} = \begin{cases} 0, & B_{ij} = 0, \\ 255, & B_{ij} = 1. \end{cases}$$

All random draws must be mutually independent. Do not clip or otherwise modify uncorrupted pixels. Because pseudorandom-number generators differ across software environments, valid implementations using different generators may produce different realizations of X_n ; state the generator algorithm used in your report.

Center the noisy matrix using the *clean-data mean image* from Problem 0:

$$\tilde{X}_n = X_n - \mu \mathbf{1}^\top.$$

Compute: For every $K \in K_{\text{set}}$, denoise by projecting onto the subspace learned from the clean centered data:

$$\hat{X}_{n,K} = \mu \mathbf{1}^\top + Q_K Q_K^\top \tilde{X}_n.$$

Compute: For each K , compute

$$e_{\text{clean}}(K) = \|X - \hat{X}_{n,K}\|_F,$$

$$e_{\text{noisy}}(K) = \|X_n - \hat{X}_{n,K}\|_F,$$

and the constant baseline error

$$e_{\text{baseline}} = \|X - X_n\|_F.$$

Do not clip the denoised reconstructions before computing these errors.

Plot: In Fig. 7, plot all three errors versus K . Plot e_{baseline} as a horizontal reference line.

Discuss: Explain the behavior of the three curves. Identify and interpret the minimizer or near-minimizing range of $e_{\text{clean}}(K)$. Explain why a small rank can discard meaningful digit structure, whereas a larger rank can preserve progressively finer-scale variation, including some corruption.

Problem 7 (18 points): SVD Analysis of Linear Systems

For each case below, analyze the linear system

$$Gw = f.$$

Use an SVD-based method or the Moore–Penrose pseudoinverse as the primary computational tool. Do not use normal equations as the primary method. Report numerical values to at least six decimal places.

$$A = \begin{bmatrix} -2.74125009 & 2.24215689 & -0.60553211 & -0.16755625 \\ -0.34868395 & 0.29538923 & -0.45259498 & 0.50015934 \\ 2.49664208 & 0.27798324 & 2.00739274 & 0.21978030 \end{bmatrix}, \quad y_A = \begin{bmatrix} 0.61339829 \\ 0.11012282 \\ -0.06426754 \end{bmatrix}.$$

$$B = \begin{bmatrix} -2.74125009 & -0.34868395 & 2.49664208 \\ 2.24215689 & 0.29538923 & 0.27798324 \\ -0.60553211 & -0.45259498 & 2.00739274 \\ -0.16755625 & 0.50015934 & 0.21978030 \end{bmatrix},$$

$$y_B = \begin{bmatrix} 0.66761214 \\ 0.35931116 \\ 0.74289966 \\ 0.02979187 \end{bmatrix}, \quad y_{B2} = \begin{bmatrix} 0.24982762 \\ -0.45768269 \\ 0.22778277 \\ 0.63413920 \end{bmatrix}.$$

$$C = \begin{bmatrix} 0.31997336 & 0.43316234 & -0.33457014 & -0.34017903 \\ 1.12969075 & 1.52931319 & -1.18122581 & -1.20102843 \\ 0.20087760 & 0.27193705 & -0.21004138 & -0.21356262 \end{bmatrix},$$

$$y_C = \begin{bmatrix} 0.14216640 \\ 0.50192948 \\ 0.08925132 \end{bmatrix}, \quad y_{C2} = \begin{bmatrix} -1.01480112 \\ 0.41152110 \\ -0.45229071 \end{bmatrix}.$$

$$D = \begin{bmatrix} 0.07999334 & 0.28242269 & 0.05021940 \\ 0.10829058 & 0.38232830 & 0.06798426 \\ -0.08364254 & -0.29530645 & -0.05251035 \\ -0.08504476 & -0.30025711 & -0.05339065 \end{bmatrix},$$

$$y_D = \begin{bmatrix} 0.41615372 \\ 0.56336601 \\ -0.43513813 \\ -0.44243299 \end{bmatrix}, \quad y_{D2} = \begin{bmatrix} 0.47277025 \\ -0.64357627 \\ 1.30059591 \\ 1.42694800 \end{bmatrix}.$$

Cases:

1. $G = A, f = y_A.$
2. $G = B, f = y_B.$
3. $G = B, f = y_{B2}.$
4. $G = C, f = y_C.$
5. $G = C, f = y_{C2}.$
6. $G = D, f = y_D.$
7. $G = D, f = y_{D2}.$

For each case, let $G \in \mathbb{R}^{m \times n}$ and compute

$$G = U\Sigma V^\top, \quad \sigma_1(G) \geq \sigma_2(G) \geq \cdots \geq 0.$$

For this problem, use the relative numerical-rank tolerance

$$\tau_G = 10^{-7}\sigma_1(G), \quad r_G = \#\{i : \sigma_i(G) > \tau_G\}.$$

Report τ_G , r_G , and all singular values of G . Let U_{r_G} and V_{r_G} contain the left and right singular vectors, respectively, associated with the singular values satisfying $\sigma_i(G) > \tau_G$. The orthogonal projector onto $\text{Range}(G)$ is

$$P_{\text{Range}(G)} = U_{r_G}U_{r_G}^\top.$$

For every pseudoinverse calculation in this problem, use the truncated-SVD pseudoinverse defined by the same tolerance:

$$G_\tau^\dagger = V_{r_G}\Sigma_{r_G}^{-1}U_{r_G}^\top.$$

That is, discard every singular value satisfying

$$\sigma_i(G) \leq \tau_G.$$

Do not use a software library's default pseudoinverse cutoff unless it is explicitly set to implement this same threshold. In all subsequent expressions in this problem, interpret $G^\dagger$ as $G_\tau^\dagger$.

For the range-membership decision, use the relative tolerance

$$\eta = 10^{-7}.$$

Declare

$$f \in \text{Range}(G)$$

if and only if

$$\|f - P_{\text{Range}(G)}f\|_2 \leq \eta \max(1, \|f\|_2).$$

A residual at or below the stated tolerance is treated as zero for the purpose of determining whether an exact solution exists.

For each case, report the following:

1. The dimensions $m \times n$, all singular values, numerical rank tolerance τ_G , and numerical rank r_G of G .
2. Whether the system is overdetermined ($m > n$), square ($m = n$), or underdetermined ($m < n$).
3. The numerical entries of $P_{\text{Range}(G)}$; do not replace the matrix by a symbolic expression. Verify numerically that it is symmetric and idempotent by reporting

$$\left\| P_{\text{Range}(G)} - P_{\text{Range}(G)}^\top \right\|_F$$

and

$$\left\| P_{\text{Range}(G)}^2 - P_{\text{Range}(G)} \right\|_F.$$

4. The projected vector $P_{\text{Range}(G)}f$ and residual

$$r_f = f - P_{\text{Range}(G)}f.$$

Report $\|r_f\|_2$ and use the prescribed criterion above to determine whether $f \in \text{Range}(G)$.

5. Whether the system has no exact solution, a unique exact solution, or infinitely many exact solutions. Justify your conclusion using consistency and rank.
6. If an exact solution exists, report one solution w . If infinitely many exact solutions exist, report the minimum- ℓ_2 -norm solution

$$w^\dagger = G_\tau^\dagger f.$$

7. If no exact solution exists, report the minimum-norm least-squares solution

$$w_{\text{LS}} = G_\tau^\dagger f$$

and its residual norm $\|Gw_{\text{LS}} - f\|_2$.